Lecture plan · for the lecturer

Learning as optimisation

AI · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Frame learning as minimising empirical risk.
- Run gradient descent and SGD.
- Diagnose learning-rate problems.

Activity: Steps by hand

1. Warm-up (5 min): Show a 2-D loss contour and ask students to draw the gradient arrow at one point.
2. Derive (15 min): Derive the gradient of squared and logistic loss on the board.
3. Step (15 min): Pairs take five steps on θ^2 with $\eta = 0.1$, 0.5 and 1.2 and plot them.
4. Code (20 min): Implement mini-batch SGD and compare loss curves for two batch sizes.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- At which η did θ^2 diverge, and why?
- Why does SGD jitter?
- How will you assess curve reading?

Exit ticket

- Sketch the loss curve for a learning rate that is too big, and label it.

Try it in your next lecture
Give each pair the loss θ^2 and three learning rates; they take five steps by hand and plot them before you show any code.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Loss θ^2, $\theta = 3$, $\eta = 0.1$. Where is θ after one step?**

◯ 2.9 ◯ 0 ◯ 2.4

2 **Take two gradient steps on the loss θ^2 with $\eta = 0.25$, starting at $\theta = 8$.**

After step 1, θ = _____ . After step 2, θ = _____ .

Word bank: 6 · 4 · 2 · 0

3 **For the loss θ^2, which learning rates make gradient descent diverge? Choose all that apply.**

☐ $\eta = 0.4$ ☐ $\eta = 1.5$ ☐ $\eta = 1.1$ ☐ $\eta = 0.9$

4 **The negative gradient always points directly at the minimum of the loss.**

◯ True ◯ False

5 **Match each loss curve to its likely cause.**

Loss falls very slowly and steadily	• • • •	Small mini-batches
Loss grows, then becomes nan	• • • •	Learning rate too small
Loss falls but jitters up and down	• • • •	Overfitting
Training loss low, validation loss rising	• • • •	Learning rate too large

Answers · for the lecturer

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1 **2.4**

The gradient is $2\theta = 6$, so $\theta = 3 - 0.1 \times 6 = 2.4$.

2 **After step 1, θ = 4. After step 2, θ = 2.**

The update is $\theta \leftarrow (1 - 2 \times 0.25)\theta = 0.5\theta$, so $8 \rightarrow 4 \rightarrow 2$.

3 **$\eta = 1.5$; $\eta = 1.1$**

$\theta \leftarrow (1 - 2\eta)\theta$ diverges when $|1 - 2\eta| > 1$, i.e. $\eta > 1$; at $\eta = 0.9$ it oscillates but still converges.

4 **False**

It points in the direction of steepest local descent; on a long narrow valley that is nearly across the valley, not along it.

5 **Loss falls very slowly and steadily — Learning rate too small**
Loss grows, then becomes nan — Learning rate too large
Loss falls but jitters up and down — Small mini-batches
Training loss low, validation loss rising — Overfitting

A slow, steady fall means steps that are too small; a loss that grows to nan means steps that overshoot; jitter comes from the noisy gradients of small mini-batches; and low training loss with rising validation loss means the model is fitting noise.

For you to think about

Where do your students first meet gradient descent? What could they do by hand before they see code?